HBSM: Hybrid Boundary Sensitivity Mechanisms

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. HBSM tests whether sensitivity in frontier hybrid reasoners can be explained and predicted cheaply. The broad forward-only sensitivity claim is wounded by KL Lens-style forward-sensitivity work, so no novelty or positive result is claimed yet.

1 Current Gate

The branch must pass sensitivity heterogeneity, no-forward predictor, and mechanism gates. If it overlaps with HORN, it should be folded into a unified hybrid-quantization paper. B1 must first reproduce sensitivity heterogeneity on current hybrid reasoners, B2 must beat no-forward baselines without overfitting depth or size, and B3 must add mechanism beyond HORN's boundary-direction test.

2 Mac Artifact Status

A deterministic synthetic B1/B2 packet validates the layer-sensitivity and cheap predictor readout. It reports kurtosis-versus-sensitivity Spearman rho 0.657 and two boundary-flagged top-decile hits, producing `SYNTHETIC_PASS_REAL_LAYER_SENSITIVITY_NEXT`. This is synthetic-only: it is not model evidence, not a novelty claim, and not GPU evidence. The metadata-only model eligibility packet identifies the current live hybrid targets and architecture hashes, but no corresponding weights are cached repo-locally. HBSM therefore has no real forward-sensitivity evidence on this Mac yet.

3 Next Evidence

The next admissible row is a real layer-sensitivity sweep on current hybrid reasoners. If cheap predictors fail to correlate with forward sensitivity, HBSM should be killed or folded into HORN rather than kept as a separate paper.

4 Admissible Real Packet

The real B1 packet must use fixed boundary flags from the shared architecture maps. Each row must report `model_id`, `layer`, `boundary_flag`, `precision_perturbation`, `kl_or_nll_drift`, `cheap_predictor`, `parameter_count`, `weight_norm`, `top_decile_flag`, `random_top_decile`, `train_test_split`, and `control_type`. The packet must include `prompt_ids_hash`, `architecture_map_hash`, `summary.md`, matching `row_count`, and pass `experimental.shared.check_gate_packet --mode real --project hbsm`. The checker requires both true and false boundary flags, finite sensitivity and predictor fields, all listed controls, and a perturbation-off row whose drift is near zero. It also requires matched top-decile/random-flag counts and both train and test split rows unless the packet records an explicit resource limit. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The summary must include top-decile count, random top-decile count, train/test counts, and boundary top-decile enrichment, Fisher p-value, and cheap-predictor Spearman correlation. The checker recomputes these fields from raw rows and rejects stale or hand-filled summaries.

Gate	Promotion threshold
B1 sensitivity	Primary boundary-only rows show top-decile enrichment over non-boundary rows with Fisher $p < 0.05$, while matched random flags do not reproduce the enrichment.
B2 predictor	Spearman $\rho \geq 0.6$ with bootstrap lower bound ≥ 0.3.
B3 mechanism	Matched-noise attention drift exceeds SSM drift and aligns with HORN.

Table 1: Pre-registered HBSM gate thresholds before any standalone claim.

Required baseline	Purpose
Random top-decile flags	Tests enrichment beyond chance.
Layer-index baseline	Controls depth effects.
Parameter-count baseline	Controls layer size.
Weight-norm baseline	Controls simple magnitude predictors.
Boundary-only baseline	Tests whether HORN already explains the effect.
Perturbation-off row	Catches hook and scoring bugs.
Train/test split	Reduces cheap-predictor overfit.

Table 2: Controls required before HBSM can claim a cheaper sensitivity predictor.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/hbsm/paper/reviewer_pack.md`. It records that HBSM is not camera-ready as a standalone paper until B1–B3 separate the mechanism from existing sensitivity tools.

6 Novelty Boundary

The closest risk is KL Lens-style forward sensitivity for mixed-precision hybrid models KL Lens (2026). HBSM can survive only if it shows a cheaper no-forward predictor or a boundary mechanism on current hybrid reasoners; otherwise it should be folded into HORN or killed.

References

Jason Kong, Nilesh Prasad Pandey, Flavio Ponzina, and Tajana Rosing. A KL Lens on Quantization: Fast, Forward-Only Sensitivity for Mixed-Precision SSM-Transformer Models. arXiv:2604.13440, 2026. https://arxiv.org/abs/2604.13440.